BCAC 0029
CYBER SECURITY AND CYBER LAWS

Cryptography

Definition

Cryptography is the science of using mathematics to encrypt and decrypt data.

Phil Zimmermann

Cryptography is the art and science of keeping messages secure.

Bruce Schneier

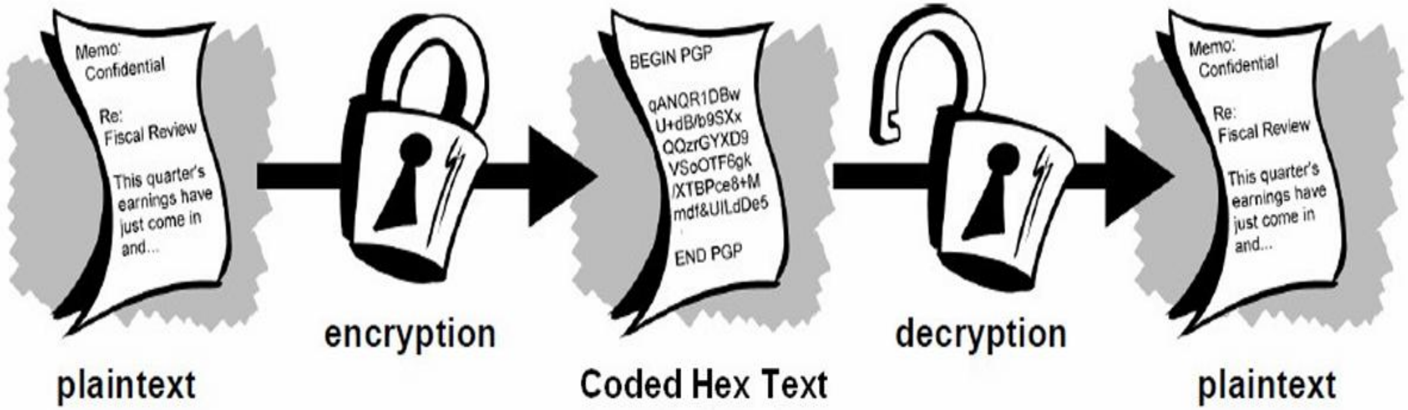
The art and science of concealing the messages to introduce secrecy in information security is recognized as cryptography.

Cryptography is the study of techniques for preventing access to sensitive data by parties who are not authorized to access the data.

Terminologies

A message is **plaintext** (sometimes called **cleartext**). The process of disguising a message in such a way as to hide its substance is **encryption**. An encrypted message is **ciphertext**. The process of turning ciphertext back into plaintext is **decryption**.

A **cipher** (or **cypher**) is an algorithm for performing encryption or decryption—a series of well-defined steps that can be followed as a



Terminology

A **cryptosystem** is an implementation of cryptographic techniques and their accompanying infrastructure to provide information security services. A cryptosystem is also referred to as a cipher system. The various components of a basic cryptosystem are as follows –

- Plaintext
- Encryption Algorithm
- Ciphertext
- Decryption Algorithm
- Encryption Key
- Decryption Key

Cryptanalysis

- Cryptanalysis is the practice of analyzing and breaking encryption systems to decipher encrypted messages (ciphertext) without the secret key.
- Often by finding weaknesses in algorithms or implementations, using techniques from mathematics, statistics, and computer science.

Types of Attacks

- **Ciphertext-Only Attack (COA):** Attacker only has ciphertext; most difficult.
- **Known-Plaintext Attack (KPA):** Attacker has ciphertext AND corresponding plaintext/ciphertext pairs.
- **Chosen-Plaintext Attack (CPA):** Attacker chooses plaintexts to encrypt and sees the ciphertext.
- **Chosen-Ciphertext Attack (CCA):** Attacker chooses ciphertexts to decrypt and sees the plaintext.

Methods & Examples

- **Frequency Analysis:** Exploiting letter/symbol frequency in simpler ciphers.
- **Linear & Differential Cryptanalysis:** Advanced mathematical methods for complex ciphers.
- **Brute-Force:** Trying every possible key.

Security Algorithms

- **Symmetric-Key Algorithms:** Use the same key for encryption and decryption. They are generally fast and suitable for large data volumes.
    - **AES (Advanced Encryption Standard):** The current standard, using 128, 192, or 256-bit keys.
    - **Blowfish/Twofish:** Fast block ciphers.
    - **DES/3DES:** Older, largely replaced by AES due to shorter key lengths.
- **Asymmetric-Key Algorithms (Public-Key):** Use a public key for encryption and a private key for decryption, solving the problem of key distribution.
    - **RSA (Rivest-Shamir-Adleman):** Uses large prime numbers for secure communication.
    - **ECC (Elliptic Curve Cryptography):** Provides higher security with smaller keys, making
    - it efficient for mobile devices.
- **Hashing Algorithms (Data Integrity):** Convert data into a fixed-size, unique string (hash) to ensure data has not been altered.
    - **SHA (Secure Hashing Algorithm):** E.g., SHA-256 is used for securing data and digital certificates.
    - **MD5:** Older, often used for checksums, but less secure.

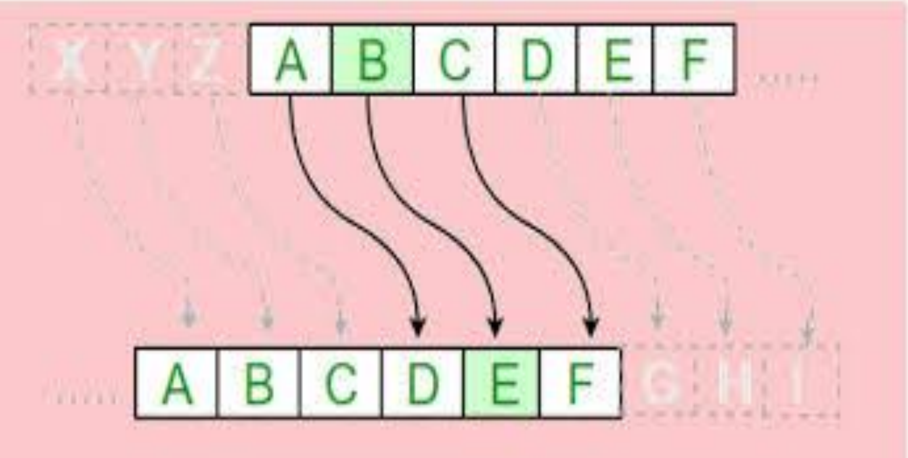
Ciphers

- Ciphers transform readable text into unreadable ciphertext, broadly categorized into Substitution (replacing letters/bits, e.g., Caesar, Vigenère) and Transposition (rearranging letters/bits, e.g., Rail Fence, Columnar).
- **Substitution ciphers**

  Replace plaintext characters with others.
- **Caesar Cipher**: Shift letters by a fixed number (e.g., A becomes D).
- **Monoalphabetic**: One-to-one mapping (e.g., A=Q, B=X).
- **Transposition Ciphers**:
- Rearrange the order of characters.
- **Rail Fence Cipher:** Write text diagonally ‘rails’ and reads off in rows.
- **Column transposition:** Writes text in columns reads by keyword determined column order.

Caesar Cipher

- The Caesar Cipher is an ancient, simple substitution cipher where each letter in plaintext is replaced by a letter fixed positions down or up the alphabet (e.g., with a shift of 3, 'A' becomes 'D'). Used by Julius Caesar for secure communication, it is easy to use but weak, as it is easily broken via brute force (only 25 possible keys) or frequency analysis.
- **Example**
- **Plain text: ABC**
- **Key =3**
- **Ciper Text = DEF**



Transposition Cipher

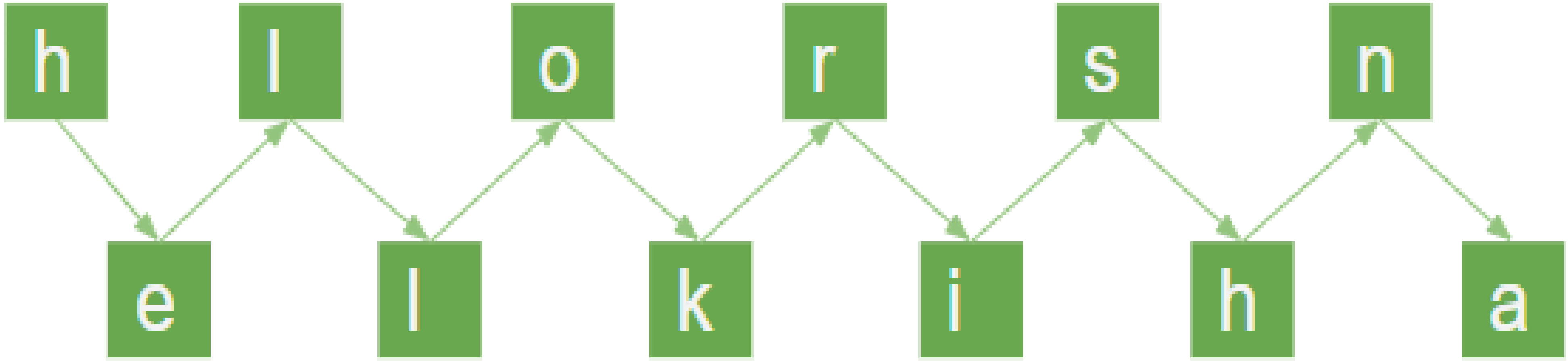
- Transposition ciphers are cryptographic methods that rearrange the order of characters in a plaintext message without substituting them.

- **Types of Transposition cipher-**
- Rail Fence (zigzag pattern),
- Columnar Transposition (key-based grid reordering),
- Double Transposition (two-layer security),

Rail Fence Transposition Cipher

- Rail Fence Transposition cipher technique is the simplest transposition cipher technique. It is also termed as a zigzag cipher.

Example: The plain text is "Hello Krishna“ by assuming rail size= 2
Now, we will write this plain text in the diagonal form:



Block (Single Columnar) Transposition Cipher

- In this technique, first, we write the message or plaintext in rows. After that, we read the message column by column. In this technique, we use a keyword to determine the no of rows.
- Step 1: First we write the message in the form of rows and columns, and read the message column by column.
- Step 2: Given a keyword, which we will use to fix the number of rows.
- Step 3: If any space is spared, it is filled with null or left blank or in by (_).
- Step 4: The message is read in the order as specified by the keyword.

S	E	C	R	E	T
4	1	0	3	2	5
T	H	I	S	I	S
A	P	L	A	I	N
N	T	E	X	T	F
O	R	D	E	M	O
N	S	T	R	A	T
I	O	N	X	X	X

Key

Plain Text: This is a plain text to demonstrate the use of the transposition cipher.

Cipher Text: ILEDTN HPTRSO IITMAX SAXERX TANONI SNFOTX

One Time Pad(Vernam Cipher)

- The OTP is classical encryption technique that provides perfect secrecy, meaning it is theoretically unbreakable if use correctly.
- A OTP is know as symmetric encryption technique where:

-- The plaintext is encrypted by combining it with a random key of the same length.

--- Each character of the plaintext is combined with a character from the key using modular arithmetic.

- **Step 1: Convert letters to Numbers**
- **Encryption Formula C= (P+K)mod 26**
- **Decryption Formula P= (C- K+26)mod 26**

Example: for Encryption

- **Plaintext: HELLO**
- **Keyword: XMCKL**

Plaintext	Position (0–25)	Key	Key Position (0–25)	C = (P+K)%26	Ciphertext
H	7	X	23	(7+23)%26=4	E
E	4	M	12	(4+12)%26=16	Q
L	11	C	2	(11+2)%26=13	N
L	11	K	10	(11+10)%26=21	V
O	14	L	11	(14+11)%26=25	Z

Decryption

Decryption Formula P= (C- K+26)mod 26

- **Ciphertext: EQNVZ**
- **Keyword: XMCKL**

Ciphertext	Position (0–25)	Key	Key Position (0–25)	P = (C - K + 26) % 26	Plaintext
E	4	X	23	(4 - 23 + 26)%26 = 7	H
Q	16	M	12	(16 - 12 + 26)%26 = 4	E
N	13	C	2	(13 - 2 + 26)%26 = 11	L
V	21	K	10	(21 - 10 + 26)%26 = 11	L
Z	25	L	11	(25 - 11 + 26)%26 = 14	O

Vernam Cipher use of XOR:

- The most famous use of XOR in cryptography is in the one-time pad, also known as the Vernam Cipher.
- This system works on binary data (bits) rather than letters.
- In this system, a plaintext message is XORed with a truly random key or pad that is as long as the message itself.
- The technique can be expressed as follows: **Ci =** $\boldsymbol{P}$**i** $\oplus$ $\boldsymbol{K}$**i**
- The result is a completely random ciphertext, which is theoretically unbreakable if the key is truly random, used only once, and kept secret.
- The recipient, who has the same key, can XOR the ciphertext with the key to retrieve the original message. **Decryption Pi =** $\boldsymbol{C}$**i** $\oplus$ $\boldsymbol{K}$**i**

Example

- **Encryption: $Ci = Pi \oplus Ki$,**
- **Decryption $Pi = Ci \oplus Ki$**
- **Msg :**
- C= 01000011
- H = 01001000
- U = 01010101
- C = 01000011
- K = 01001011

- **Key = 10101010**

Arbitrated protocol

- An arbitrated protocol is a communication or cryptographic process involving a trusted third party (the arbitrator/arbiter) to facilitate, verify, or validate transactions between two or more parties.
- Unlike self-enforcing protocols, these rely on the arbiter to resolve disputes, ensure impartiality, and manage secure exchanges.

- **Types of Arbitrated protocol**
- **Arbitrated Digital Signatures:** A sender sends messages through an arbiter who validates, timestamps, and signs the message before passing it to the receiver, enhancing security and non-repudiation.
- **Dispute/Adjudicated Protocols:** The arbiter is involved only if a conflict occurs, making it a more cost-effective, two-stage process (direct communication first, then intervention if needed).
- **CAN Bus Arbitration:** In hardware, this prevents data collision on shared networks by having nodes with higher-priority messages "win" control of the bus.

Advantages and Disadvantages

- **Advantages:**
- High impartiality,
- legal recognition of electronic signatures,
- and secure conflict resolution.
- **Disadvantages:**
- Creates a potential bottleneck,
- involves high maintenance costs for the arbitrator,
- and introduces dependency on a third party.